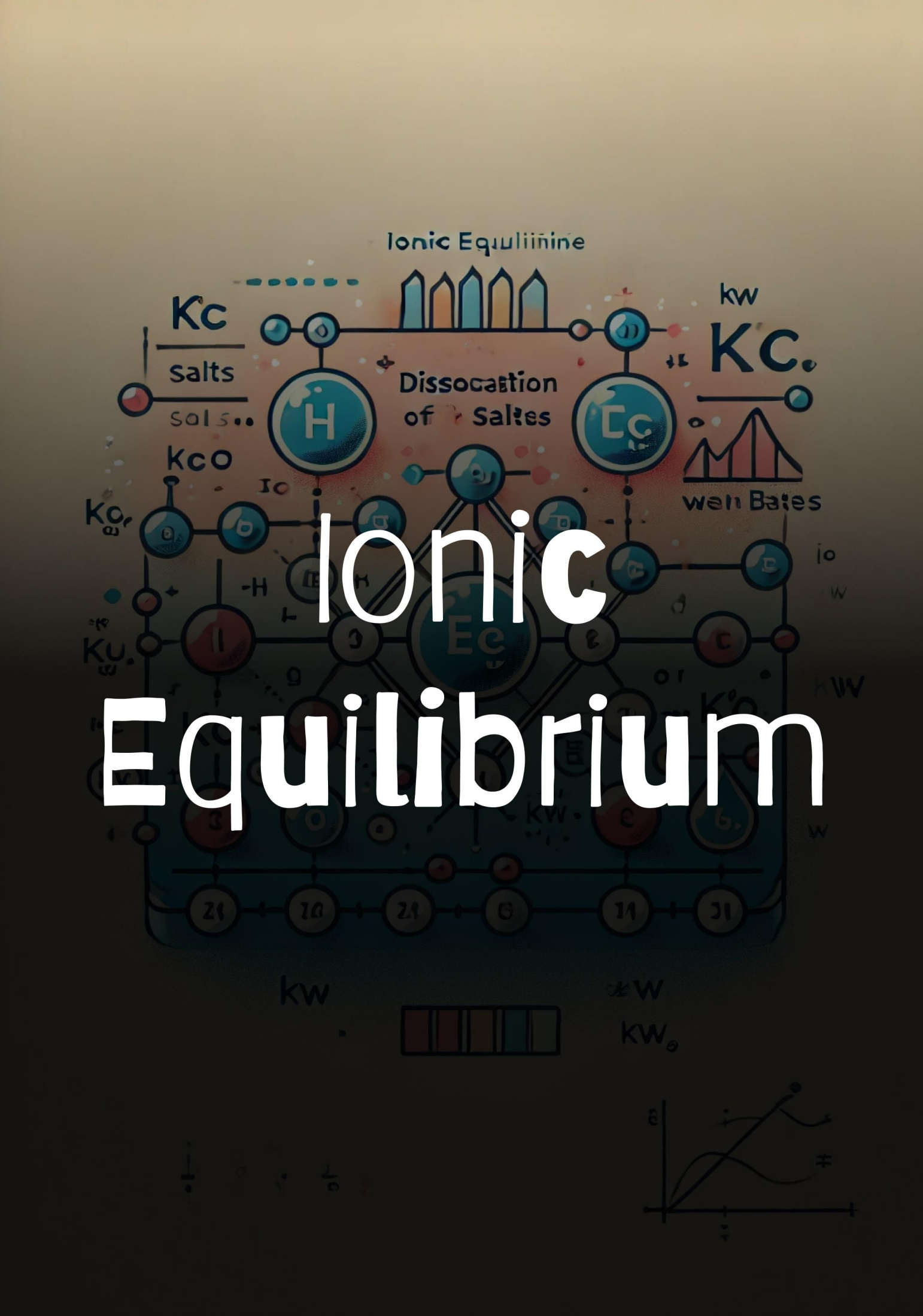

Ionic Equilibrium
Kc
salts
Dissociation
of Salts
kw
Kc
Ionic Equilibrium

IONIC EQUILIBRIUM

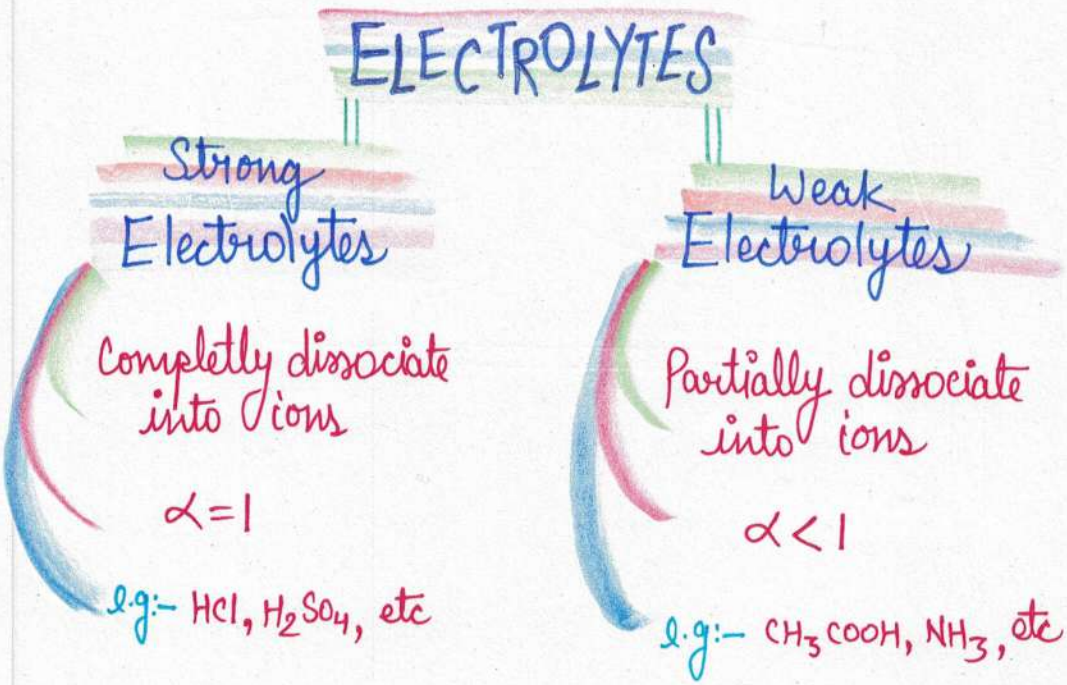
In this chapter, We'll study about equilibrium of weak electrolytes.

There are 2 types of substances.

Non electrolyte – which does not ionise in their aqueous solution.

e.g:– Urea, glucose, sucrose etc.

Electrolyte – which will dissociate into ions in their aqueous solution.



We will frequently use an operator in this chapter $\equiv P$

$P \equiv -\log$

e.g:- $-\log [H^+] = pH$ $\quad -\log [OH] = pOH$

$-\log [K_a] = pK_a$ $\quad -\log [K_b] = pK_b$

BASICS OF LOGARITHM

- $\log m^n = n \log m$
- $\log \frac{m}{n} = \log m - \log n$
- $\ln x = (\log_e x)\, 2.303$
- $\log 10 = 1$
- $\log \frac{1}{x} = -\log x$
- $\log m^n = \log m + \log n$
- $\log_a m = \frac{\log_b m}{\log_b a}$
- $\log 1 = 0$
- $\log 2 = 0.3$
- $\log 3 = 0.48$

- if $\log_a x = Y$

  $\text{antilog}\, x = a^Y$

QUESTION Calculate the following

1. $\log 25$
2. $\log 1.8$
3. $\log \frac{21}{7}$
4. $\log 5$
5. $\log \frac{22}{3}$

Solution

1. $\log 25 = \log (5)^2 = 2(\log 10/2)$

$2[\log 10 - \log 2]$

$2[1 - 0.3] = 2(0.7) = 1.4$

2. $\log 1.8 = \log \frac{18}{10} = \log 18 - \log 10$

$= \log 3^2 \times 2 - 1$

$= 2(0.48) + 0.3 - 1$

$= 0.96 + 0.3 - 1 = 1.26 - 1 \Rightarrow 0.26$

3. $\log \frac{21}{7} = \log 3 \Rightarrow 0.48$

4. $\log 5 = \log \frac{10}{2} = 1 - 0.3 \Rightarrow 0.7$

5. $\log \frac{22}{3} = \log 22 - \log 3$

$= \log 2 + \log 11 - 0.48$

$= 0.3 + \frac{\log 10 + \log 12}{2} - 0.48$

$= -0.18 +$

$\Rightarrow 0.86$

Qu:-

1. $\log 1.8 \times 10^{-5}$
2. $\log 14 \times 10^{-6}$
3. $\log 7 \times 10^{-5}$

Ans:-

1. $\log 1.8 \times 10^{-5} = 0.26 - 5 = -4.74$
2. $\log 14 - 6 = \log 7 + 0.3 - 6$

$= 0.84 + 0.3 - 6$

$= 1.14 - 6 \Rightarrow -4.86$

3. $\log 7 \times 10^{-5} = \log(7 - 5) = 0.84 - 5$

$\Rightarrow -4.16$

REMEMBER

$\log 1.8 = 0.26$

$\log 7 = 0.84$

$\log 5 = 0.7$

$\log 2 = 0.3$

$\log 3 = 0.48$

ACID BASE CONCEPT

1. ARRHENIUS THEORY

According to Arrhenius, substances that provide H^+ ion in their aqueous solution are acids and substances which provide OH^- ion in their aqueous solution are base.

For eg:- $\underset{\text{acid}}{HCl} \rightleftharpoons H^+ + Cl^-$ $\qquad$ $\underset{\text{base}}{NaOH} \rightleftharpoons Na^+ + OH^-$

2. BRONSTED AND LOWRY THEORY

Proton acceptors are bases and proton donors are acid.

$\underset{\text{base}}{B} + H^+ \longrightarrow \underset{\text{Conjugate acid}}{BH^+}$

$\underset{\text{acid}}{HA} \rightarrow H^+ + \underset{\text{Conjugate base}}{A^-}$

for eg:- $\underset{\text{acid}}{H_2SO_4} \longrightarrow H^+ + \underset{\text{Conjugate base}}{HSO_4^{2-}}$

$\underset{\text{acid}}{HSO_4^-} \longrightarrow H^+ + \underset{\text{conjugate base}}{SO_4^{2-}}$

but $\underset{\text{acid}}{H_2SO_4} \longrightarrow 2H^+ + \underset{\text{Not Conjugate base}}{SO_4^{2-}}$

$H_3PO_4 \longrightarrow H^+ + H_2PO_4^-$

$H_2PO_4 \longrightarrow H^+ + HPO_4^{2-}$

$HPO_4^{2-} \longrightarrow H^+ + PO_4^{3-}$

but $\underbrace{H_3PO_4, H_2PO_4^-, HPO_4^{2-}}_{\text{acids}}$ $\qquad$ $\underbrace{H_2PO_4^-, HPO_4^{2-}, PO_4^{3-}}_{\text{Conjugate base}}$

$\underset{\text{acid}}{H_3PO_4} \longrightarrow 3H^+ + \underset{\text{Not Conjugate base}}{PO_4^{3-}}$

3. LEWIS THEORY

Lone pair donors are **base**.
Lone pair acceptors are **acids**.

$$H_3N: \longrightarrow BF_3$$

▷ According to first two theories, water is a neutral solution but according to third theory H_2O is Lenius base.

PROPERTIES OF WATER

1. It is a Neutral Solution.
2. It is a pure liquid having concentration $= \frac{1000}{18} = 55.55\,M = 55.55$
3. At 25°C H_2O dissociates into H^+ and OH^- ion having concentration $10^{-7}M$ each.

$$\underset{(l)}{H_2O} \rightleftharpoons \underset{10^{-7}M}{H^+} + \underset{10^{-7}M}{OH^-} \qquad \Delta H > 0$$

4. K_w = ionic product of water

$$K_w = [H^+][OH^-]$$

taking -log on both sides

$$-\log K_w = -\log [H^+][OH^-]$$

$$pK_w = -\log [H^+] - \log [OH^-]$$

$$\boxed{pK_w = pH + pOH}$$

at 25°C $pK_w = pH^+ + pOH = 14$

We can predict the nature of solution by pH or pOH

@. $[H^+] = [OH^-]$ Natural Solution

$-\log[H^+] = -\log[OH^-]$

$pH + pH = pOH + pH$

$2pH = 14$

$pH = 7$

b. $[H^+] > [OH^-]$ Acidic solution

$-\log[H^+] < -\log[OH^-]$

$pH + pH < pH + pOH$

$2pH < 14$

$pH = 7$

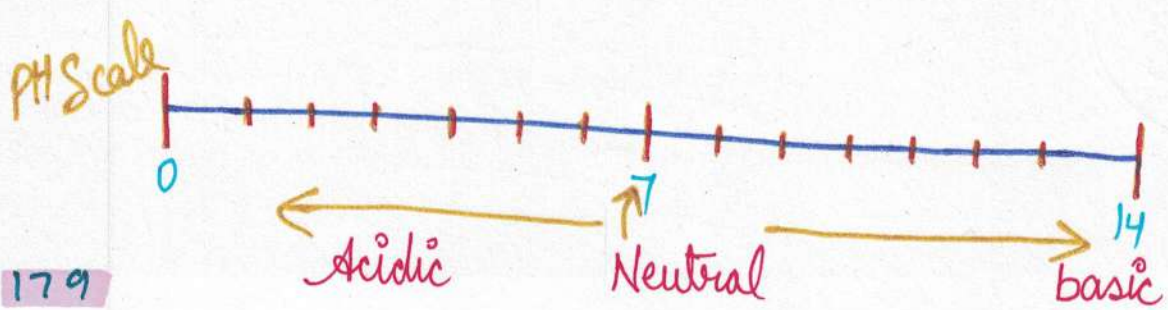
$[H^+] < [OH^-]$ Basic Solution

$-\log[H^+] > -\log[OH^-]$

$pH + pH > pOH + pH$

$2pH > 14$

$pH > 7$



Que. Predict the Nature of soln by looking at pH values if at 90°C $K_w = 10^{-13}$ for water.

(a) pH = 7 (b) pH = 6 (c) pH = 6.5 (d) pH = 8

Soln:- $pK_w = pH + pOH \Rightarrow 13$

Neutral soln if $[H^+] = [OH^-]$

$$2pH = pH + pOH$$

$$pH = \frac{13}{2} = 6.5$$

acidic soln if $pH < 6.5$

basic soln if $pH > 6.5$

So now, @ $pH > 6.5 \Rightarrow$ basic soln

pH = 6 (b) $pH < 6.5 \Rightarrow$ Neutral soln

(c) $pH = 6.5 \Rightarrow$ Neutral soln

pH = 8 (d) $pH > 6.5 \Rightarrow$ basic soln

Q. What will be effect on pH and pOH of solution if temparature will increase ?

Soln:- On increasing temparature, K_w will increase hence, both $[H^+]$ and $[OH^-]$ if increase, so pH and pOH will both decrease.

5. Degree of Dissaciation of H_2O at 25°C

$$H_2O \rightleftharpoons H^+ + OH^-$$

t = 0	c			
t = t	c-x	x	x	

$\alpha = \frac{x}{c}$

$$\alpha = \frac{10^{-7}}{\frac{1000}{18}} = 1.8 \times 10^{-9}$$

$\% \alpha = 1.8 \times 10^{-7}$

ABSOLUTE DISSOCIATION CONSTANT
OR
IONISATION CONSTANT FOR WATER

$$H_2O \rightleftharpoons H^+ + OH^-$$

K_a = dissociation constant for weak acid

K_b = dissociation constant for weak base

at 25°C $K_a = K_b = \dfrac{[H^+][OH^-]}{[H_2O]}$

$$= \frac{10^{-7} \times 10^{-7}}{\frac{1000}{18}} = 1.8 \times 10^{-16}$$

$$-\log K_a = -\log K_b = -\log 1.8 \times 10^{-16}$$

$$pK_a = pK_b = -[0.26 - 16.00]$$

$$pK_a = pK_b = -[-15.74]$$

$$pK_a = pK_b = 15.74$$

On increasing temperature K_a K_b will increase pK_a, pK_b will decrease.

$$HA \rightleftharpoons H^+ + A^+$$

$$K_a = \frac{[H^+][A^-]}{[HA]}$$

$$A^-_{(aq)} + H_2O_{(l)} \rightleftharpoons HA_{(aq)} + OH^-_{(aq)}$$

$$K_b = \frac{[HA][OH^-]}{[A^-]}$$

$$K_a K_b = \frac{[H^+][A^-]}{[HA]} \cdot \frac{[HA][OH^-]}{[A^-]} \Rightarrow K_a K_b = [H^+][OH^-]$$

$$K_a K_b = 10^{-14}$$

$$-\log K_a - \log K_b = -\log 10^{-14}$$

$$pK_a + pK_b = 14$$

$K_a K_b$ = Constant = 10^{-14}

∵ $K_a K_b$ is constant, if acid is strong, then it conjugate base should be weak.

PH Calculation

Calculate pH of :-

1. 300 ml 0.1 M HCL

$[H^+] = \frac{0.1}{300} \times 1000$

$-\log(1/3) = \log 3$

$pH = 0.48$

300 ml = 0.1

1000 ml = $\frac{0.1}{300} \times 1000$

$= 1/3$ N

$HCL \longrightarrow H^+ + Cl^-$

$[H^+] = 0.1M$

$pH = -\log 0.1$

$pH = 1$

2. 500 ml 0.1M HCl

$PH = 1$

3. 300 ml 0.2 M HCl + 200 ml 0.4 M H_2SO_4

$[H^+]$ from HCL = 0.2 $[H^+]$ from H_2SO_4 = 0.8

Total $[H^+] = 1$

$-\log[H^+] = -\log 1$

millimole of H^+ from HCl = $300 \times 0.2 = 60$

H^+ from $H_2SO_4 = 0.8 \times 200 = 160$

$\log 11 = \frac{\log 12 + \log 10}{2}$

$= \frac{2(0.3) + 0.48 + 1}{2}$

$\log \frac{22}{5} = \frac{1.08+1}{2} \Rightarrow \frac{2.08}{2}$

$-\log 5$

$= 0.3 + 1.04 - [1 - 0.3]$

$= 1.34 - (0.7)$

$= 0.64$

Total = 160 + 60 = 220 millimole

$[H^+] = \frac{22\phi}{50\phi} = \frac{22}{50}$

$= -\log \frac{22}{50}$

$= -\left[\log \frac{22}{50} - \log 10\right]$

$= -\left[\log \frac{22}{5} - 1\right]$

$= -(0.64 - 1)$

PH = .36

pH = .36

➢ 300 ml 0.2 M H_2SO_4 + 300 ml 0.2 M HCl + 200 ml 0.2M HNO_3

mill. from H_2SO_4 = 0.2 × 300 × 2

= 120

from HCL = 0.2 × 300 = 60

from HNO_3 = 0.2 × 200 = 40

Total = $[H^+] = \frac{120 + 60 + 40}{800} = \frac{22\phi}{80\phi}$

$\log 11 + \log 2$

$= \log 11 \times 2$

$\log 22 = 1.34$

$-\log [H^+] = -\log \left(\frac{22}{80}\right)$

$= -[\log 2 + \log 11 - \log 8 - 1]$

$= -[1.34 - 0.9 - 1]$

$= -[0.44 - 1]$

$= 0.56$

4. 300 ml 0.1M NaOH

$[H^+] = 0$ $\quad pOH = 1$

$pH = 14 - 1$

pH = 13

5. 300 ml 0.2 M NaOH + 200 ml 0.2 M $Ca(OH)_2$

$$[OH^-] = \frac{60+80}{500} = \frac{140}{500}$$

$$-\log[OH^-] = -\log\left[\frac{14}{50}\right]$$

$$= -[+\log 7 + \log 2 - 2\log 5 - \log 2]$$

$$-(.84 + 0.3 - 2(1-0.3) - 0.3)$$

$$-(.84 - 1.4)$$

$$= 0.56 \Rightarrow pH = 14 - 0.56 \Rightarrow 13.44$$

$\boxed{pH = 13.44}$

6. 300 ml 0.2 M $Ca(OH)_2$ + 200 ml 0.2 M NaOH + 300 ml 0.1 M $Al(OH)_3$

$$[OH] = \frac{120+40+90}{800} = \frac{250}{800}$$

$$pOH = -\log\frac{25}{80}$$

$$= -[2\log 5 - 3\log 2]$$

$$= -[2(0.7) - 3(0.3)]$$

$$= -[1.4 - 0.9] = 0.5$$

$$pH = 14 - 0.5 \Rightarrow 13.5$$

$\boxed{pH = 13.5}$

Acid + Base $\longrightarrow$ Salt + Water

$$H^+ + OH^- \rightarrow H_2O$$

millimole of H^+ = millimole of OH^- $\longrightarrow$ neutral

millimole of H^+ > millimole of OH^- $\longrightarrow$ acidic

millimole of H^+ < millimole of OH^- $\longrightarrow$ basic

Que:- 300 ml 0.2M NaOH + 200 ml 0.3M HCL

millimole $[OH^-] = 0.2 \times 300 = 60$

millimole $[H^+] = 0.3 \times 200 = 60$

millimole $[H^+]$ = millimole OH^-

pH = 7

Que:- 300 ml 0.3M HCL + 200 ml 0.2 M NaOH

millimole $H^+ = 90$ $\quad OH^- = 40$

$K_w = [OH^-][H^+] = 0.06$ $\quad H^+ > OH^-$ $\quad pH$ $H^+ = \frac{56}{500}$ millimole.

$pH = -\log \frac{6}{100} = -(0.78 - 2)$ acidic $\quad 1/10$

$pH = 1$

Que:- 300 ml 0.2 M HCl + 200 ml 0.5M NaOH

milli $H^+ = 60$ $\quad$ milli $OH^- = 100$

$OH^- > H^+$

basic $\quad pH > 7$

$pH = 1.1$

pH = 12.9

Que:- What is the pH of resulting soln if pH=2 and pH=4 are added in following volume ratio :-

(a) 1:1 $\quad$ (b) 2:3 $\quad$ (c) 3:2

Soln:- $2 = -\log H^+$

$10^{-2} = [H^+]$ $\quad [OH^-] = 10^{-12}$

$10^{-2} = \frac{n_1}{V_1}$

$4 = -\log H^{-1}$

$10^{-4} = [H^+]$

$10^{-10} = [OH^-]$

$10^{-4} = \frac{H_2}{V_2}$

$n_1 : H_2 = \frac{10^{-2}}{10^{-4}} = 10^2$ $\quad V_2 = \frac{H_2}{10^{-4}}$

$$[H^+] = \frac{n_1 + n_2}{\frac{2n_2}{10^{-4}}} = \frac{10^2 n_2 + n_2}{\frac{2n_2}{10^{-4}}} = \frac{10/n_2}{2n^2/10^{-4}}$$

$$[H^+] = \frac{101 \times 10^{-4}}{2}$$

$$-pH = -(\log 101 - 4 - 0.3)$$

$$= -(2 - 4.3)$$

$$\boxed{pH = 2.3}$$

2:3

$$n_1 : n_2 = \frac{n_1 + n_2}{2n_2} \quad 2:3$$

$$\frac{V_1}{V_2} = \frac{2}{3} \qquad \frac{n_1}{10^{-2}} \times 10^{-4-2} = \frac{2}{3}$$

$$\frac{n_1}{V_1} = 10^{-2} \qquad \frac{n_1}{n_2} = \frac{2}{3} \times 10^2 = \frac{200}{3}$$

$$n_1 = \frac{200}{3} n_2$$

$$[H^+] = \frac{10^{-2} V_1 + 10^{-4} V_2}{5V_1 + V_2}$$

$$[H^+] = \frac{2V \times 10^{-2} + 3V \times 10^{-4}}{2V + 3V}$$

$$= \frac{2 \times 10^{-2} + 3 \times 10^{-4}}{5}$$

$$[H^+] = \frac{0.02}{10} + \frac{0.0003}{5} = \frac{0.0203}{5}$$

$$= -\log\left(\frac{0.0205}{5}\right)$$

$$= -\left(\log \frac{2050}{10000} - 0.7\right)$$

$$= -(\log 200 - 4 - 0.7)$$

$$= -(\log 2 + 2 - 4.7)$$

$$= -(2.3 - 4.7)$$

$$\boxed{pH = 2.4}$$

3V 10^{-2} H 2V 10^{-4}

$$[H^+] = \frac{0.03 + 0.0002}{5}$$

$$= \frac{0.0302}{5} \simeq \frac{0.0300}{5}$$

$$= -\log\left(\frac{0.03}{5}\right)$$

$$= -(\log 3 - 2 - 0.7)$$

$$= -(0.48 - 2.70)$$

$$= -(-2.22)$$

$$= 2.22$$

Que:- $pH = 12$ $pH = 10$

$[H^+] = 10^{-12}$ $[H^+] = 10^{-10}$

(i) V 10^{-12} M V 10^{-10} M

$$[H^+] = \frac{10^{-12} + 10^{-10}}{2} = \frac{0.01 \times 10^{-10} + 10^{-10}}{2}$$

$$[H^+] = \frac{1.01 \times 10^{-10}}{2}$$

$$pH^+ = (10 - 0.3)$$

$$= -(9.7)$$

$$\boxed{pH^+ = 9.7}$$

$pOH = 2$ $pOH = 4$

$[OH^-] = 10^{-2}$ $[OH] = 10^{-4}$

(ii 2V 10^{-2} M 3V 10^{-4}

$$[OH^-] = \frac{0.02 + 0.0003}{5} = \frac{0.0200}{5}$$

$pOH = 2.4$ $pH = 14 - 2.4 = 11.6$

(iii) $pH = 2.2$

$pOH = 14 - 2.22 = 11.78$

Qu:- $pH = 12$ $\qquad$ $pH = 2$

$pOH = 2$ $\qquad$ $pH = 2$

$[OH^-] = 10^{-2}$ $\qquad$ $[H^+] = 10^{-2}$

$[H^+] = 10^{-12}$

$2V\ 10^{-12}M$ $\qquad$ $3V\ 10^{-2}M$

$$[H^+] = \frac{2\times10^{-12} + 3\times10^{-2}}{5}$$

$$= \frac{2\cdot X 0.00 \text{ --- } + 0.03}{5} \simeq \frac{0.03}{5}$$

$$pH = \left[-\log\frac{0.03}{5}\right] = -(0.48 - 2 - 0.7)$$

$$= -(-2-0.7) + 0.48$$

$[OH^-] = 10^{-2}$ $\qquad$ $[H^+] = 10^{-2}$

$10^{-2} = n/V_1$ $\qquad$ $10^{-2} = n/V_2$

$10^{-2} = 3n/2V_2$ $\qquad$ $10^{-2} = n/V_2$

$n = \frac{2V_2}{3}\times10^{-2}$ $\qquad$ $n = V_2\times10^{-2}$

millimole (OH^-) $\qquad$ millimole $[H^+] \longrightarrow H_2O$

$3V\times10^{-2}$ $\qquad$ $2V\times10^{-2}$

$[OH^-]$ $V\times10^{-2}$

$$[OH^-] = \frac{10^{-2}}{5} \Rightarrow -\log\left(\frac{10^{-2}}{5}\right)$$

$$\Rightarrow -(-2-0.7)$$

$$pH = 14 - 2.7 \Rightarrow 11.3$$

$3V\ 10^{-2}\ M\ H^+$ $\quad 2V\ 10^{-2}\ M\ H^-$

$H^+ = V \times 10^{-2}$ $\quad [H^+] = \frac{10^{-2}}{5}$

$\boxed{pH = 2.7}$

p^H Calculation for Weak Acid

CH_3COOH (K_a) $\quad CH_3COOH \rightleftharpoons CH_3COO^- + H^+$

$t=0$	C		
$t=t$	$C(1-\alpha)$	$C\alpha$	$C\alpha$

$$K_a = \frac{C\alpha^2}{1-\alpha}$$

$1-\alpha \simeq 1$ $\quad \alpha = \sqrt{\frac{K_a}{C}}$

$$[H^+] = C\alpha = C\sqrt{\frac{K_a}{C}} = \sqrt{CK_a}$$

$$-\log[H^+] = pH = -\frac{1}{2}(\log C + \log K_a)$$

$$\boxed{pH = \tfrac{1}{2}(pK_a - \log C)}$$

Que:- What is pH of 0.1 M CH_3COOH $(K_a = 1.8 \times 10^{-5})$

$CH_3COOH \rightleftharpoons CH_3COO^- + H^+$

$t=0$	0.1		
$t=t$	$0.1(1-\alpha)$	0.1α	0.1α

$$K_a = \frac{C\alpha^2}{1-\alpha} \Rightarrow 1.8 \times 10^{-5} = \alpha^2 \times 0.1$$

$$\alpha = \sqrt{1.8 \times 10^{-5} \times 10}$$

$$\alpha = \sqrt{1.8 \times 10^{-4}}$$

$$\alpha = \sqrt{1.8}\ \times 10^{-2}$$

$pH = -\frac{1}{2}(0.26 - 6)$

$pH = -\frac{1}{2}(-5.74) \Rightarrow 2.87$

$\%\alpha = \sqrt{1.8 \times 10^{-2}} \times 100$

$\sqrt{1.8}\% = 1.34\%$

if $K_a < 10^{-3}$, then it is weak acid.

pH Calculation for Weak Base

NH_3 or NH_4OH (K_b)

$$NH_3 + H_2O \rightleftharpoons NH_4^+ + OH^-$$

or

$$NH_4OH \rightleftharpoons NH_4 + OH^-$$

$t=0$	c		
$t=t$	$c(1-\alpha)$	$c\alpha$	$c\alpha$

$$K_b = \frac{[OH^-][NH_4^+]}{[NH_4OH]}$$

$$K_b = \frac{(c\alpha)(c\alpha)}{c(1-\alpha)} = \frac{c\alpha^2}{1-\alpha}$$

$1-\alpha \simeq 1$

$$K_b = c\alpha^2$$

$$\alpha = \sqrt{\frac{K_b}{c}}$$

$$[OH^-] = c\alpha = c\sqrt{\frac{K_b}{c}} \Rightarrow \sqrt{cK_b}$$

$$-\log [OH^-] = pK_b = -\tfrac{1}{2}(\log c + \log K_b)$$

$$= \tfrac{1}{2}(\log K_b - \log c)$$

Que:- What is pH of 0.1 M NH_3 $(K_b = 1.8 \times 10^{-5})$

$$NH_3 + H_2O \rightleftharpoons NH_4^+ + OH^-$$

0.1			
$(0.1-x)$		x	x

$$K_b = \frac{x^2}{0.1-x}$$

$$x = \sqrt{K_b \times 0.1}$$

$$x = \sqrt{1.8 \times 10^{-6}} \qquad x = \sqrt{1.8 \times 10^{-3}}$$

$pK_b = \frac{1}{2}(\log 1.8\times10^{-5} - \log 0.1)$

$pH = -\frac{1}{2}\left[\frac{2.6}{2} - 3\right] = 3 - 0.13 = 2.87$

$pK_b = \frac{1}{2}[0.26 - 5 + 1]$

$= \frac{1}{2}[51.74] \Rightarrow 2.87$

$pH = 14 - 2.87 \Rightarrow 11.13$

Strong acid + Weak acid

$HCl + CH_3COOH\ (K_a)$

$c_1 \qquad c_2$

$HCl \longrightarrow H^+ + Cl^-$

$c_1 \qquad c_1$

$CH_3COOH \rightleftharpoons CH_3COO^- + H^+$

$c_2 - x \qquad x \qquad c_1 + x$

$K_a = \frac{x(c_1+x)}{c_2 - x}$

$x = \frac{c_2 K_a}{c_1}$

$[H^+] = c_1 + x$

$= c_1$

Qu:- $CH_3COOH + HCl$

$K_a = 1.8\times10^{-5}$ $\quad pH = ?$ $\quad \alpha\ CH_3COOH$

$1.8\times10^{-5} = \frac{c_1 x}{c_2}$

$x = 1.8\times10^{-5} \times \frac{0.1}{0.1}$

$\alpha = \frac{x}{c_2} = \frac{1.8\times10^{-5}}{0.1} \Rightarrow 1.8\times10^{-4}$

$\% \alpha = 1.8\times10^{-2}\ \%$

$[H^+] = 0.1$

$-\log H^+ = -\log 0.1$ $\quad \boxed{pH = 1}$

dissociation of CH_3COOH is decreased further due to common ion effect.

In a soln of strong acid and weak acid pH of soln will be generated by strong acid.

Strong base + Weak Base

$NaOH + NH_3 (K_b)$

$c_1 \quad c_2$

$$NaOH \longrightarrow Na^+ + OH^-$$

$$NH_3 + H_2O \rightleftharpoons NH_4 + OH^-$$

$c_2 - x \qquad x \qquad c_1 + x$

$$K_b = \frac{[NH_4^+][OH^-]}{[NH_3]}$$

$$K_b = \frac{x(c_1 + x)}{c_2 - x} \quad (x \text{ is small})$$

$$x = \frac{c_2 K_b}{c_1}$$

$$\alpha NH_3 = \frac{x}{c_2} \qquad [OH^-] = c_1 + x \quad (x \text{ is small})$$

$$[OH^-] + c_1$$

Que.:- $NaOH + NH_3$ $\quad (K_b = 1.8 \times 10^{-5})$

$0.1\,M \quad 0.1\,M$

$[OH^-] = 0.1$

$pOH = 1 \qquad pH = 13$

$$x = \frac{c_2 K_b}{c_1} = 0.1\left(\frac{1.8 \times 10^{-5}}{0.1}\right)$$

$$\alpha = \frac{1.8 \times 10^{-5}}{0.1} = 1.8 \times 10^{-4}$$

$$\% \alpha = 1.8 \times 10^{-2} \%$$

pH for strong base + weak base will be governed by strong base.

Weak Base + Weak Acid

	$c_1 K_a$	$c' K_a$		
	CH_3COOH	CH_3COO^-	H^+	
K_{a_1}	$c_1 - x$	x	$x+y$	
	CCl_3COOH	CCl_3COO^-	H^+	
K_{a_2}	$c_2 - y$	y	$x+y$	

$$K_{a_1} = \frac{x(x+y)}{c_1 - x} \Rightarrow c K_a = x(x+y)$$

$$K_{a_2} = \frac{y(x+y)}{c_2 - y} \Rightarrow c_2 K_{a_2} = (x+y)y$$

$$c_1 K_{a_1} + c_2 K_{a_2} = (x+y)(x+y)$$

$$(x+y) = \sqrt{c_1 K_{a_1} + c_2 K_{a_2}}$$

$$[CH_3COO^-] = x = \frac{c_1 K_{a_1}}{x+y} \qquad [CCl_3COO^-] = Y = \frac{c_2 K_{a_2}}{x+y}$$

$$x = \frac{c_1 K_{a_1}}{\sqrt{c_1 K_{a_1} + c_2 K_{a_2}}} \qquad Y = \frac{c_2 K_{a_2}}{\sqrt{c_1 K_{a_1} + c_2 K_{a_2}}}$$

$$\alpha_{CH_3COOH} = x/c_1 \qquad \alpha_{CCl_3COOH} = Y/c_2$$

Qus CCl_3COOH + CH_3COOH

0.1M $K_a = 7 \times 10^{-5}$ 0.1M $K_a = 2 \times 10^{-5}$

(i) $[H^+] = \sqrt{0.1 \times 7 \times 10^{-5} + 0.1 \times 2 \times 10^{-5}}$

$\sqrt{(7+2)\,10^{-6}} = 3 \times 10^{-3}$

$-\log [H^+] = pH = -\log (3 \times 10^{-3})$

$pH = -[0.48 - 3]$

$pH = 2.52$

(ii) $[CH_3COO^-] = \frac{C_2 K a_2}{x+y} = \frac{0.1 \times 2 \times 10^{-5}}{3 \times 10^{-3}}$

$= 0.66 \times 10^{-3}$

$[CCl_3COO^-] = \frac{0.1 \times 7 \times 10^{-5}}{3 \times 10^{-3}} = 2.33 \times 10^{-3}$

(iii) $\alpha_{CH_3COOH} = \frac{x}{c_1} = \frac{0.66 \times 10^{-3}}{0.10}$

$= 6.6 \times 10^{-3}$

$\% \alpha = 0.66\%$

$\alpha_{CCl_3COOH} = 23.3 \times 10^{-3}$

$y/c_2 = 23.3 \times 10^{-3}$

$\% \alpha = 2.33\%$

Q. 200ml 0.1M HCl + 300ml 0.2M H_2SO_4 + 300ml 0.2M CH_3COOH ($Ka = 2No^-$

$[H^+] = \frac{20 + 60 \times 2}{800} = \frac{140}{800}$

$-\log H^+ = pH = -\log 14/80$

$= -[\log 7 + 0.3 - \log 80]$

$= -[1.13 - (1.9)]$

$= [-0.77] = 0.77$

$\% \alpha_{CH_3COOH} = x/c = \frac{0.2 \times 2 \times 10^{-5}}{0.3} = \frac{0.4 \times 10^{-3}}{0.3}$

$= 1.33 \times 10^{-3} \times 100$

$= 1.33 \times 10^{-1}$

$= 0.133\%$

$HCl \longrightarrow H^+ + Cl^-$

$H_2SO_4 \longrightarrow 2H^+ + SO_4^{2-}$

$CH_3COOH \longrightarrow CH_3COO^- + H^+$

0.2

0.2 − x x x

$x = \sqrt{eKa}$

$x = \sqrt{0.2 \times 2 \times 10^{-5}} = 2 \times 10^{-5}$

$CH_3COOH \rightleftharpoons CH_3COO^- + H^+$

$\frac{60}{800} - x$ $\qquad$ x $\qquad$ $\frac{140}{800} + x$

$K_a = \frac{\frac{140}{800}x}{60/800}$ $\qquad$ $x = \frac{2 \times 10^{-5} \times 6}{14_7}$

$x = \frac{6}{7} \times 10^{-5}$

$\%\alpha_{CH_3COOH} = \frac{x}{60/800} \times 100$

$= \frac{6}{7} \times \frac{10^{-5}}{60} \times 8 \Rightarrow \frac{8}{7} \times 10^{-5}$

DILUTION

$HCL \longrightarrow H^+ + Cl^-$

0·1M $\qquad$ 0·1

$H_2O \rightleftharpoons H^+ + OH^-$

$0.1+x \qquad x$

$(0.1+x)x = 10^{-14}$ $\qquad$ $[H^+] = 0.1 + 10^{-13}$

$x = 10^{-13}$ $\qquad$ $-\log[H^+] =$ **pH = 1**

$\alpha_{H_2O} = x/c = \frac{10^{-13}}{1000/18} \simeq 1.8 \times 10^{-15}$

$HCL \longrightarrow H^+ + Cl^-$

$10^{-2} \qquad 10^{-2}$

$H_2O \rightleftharpoons H^+ + OH^-$

$10^{-2}+x \qquad x$

$(10^{-2}+x)x = 10^{-14}$ $\qquad$ $[H^+] = 10^{-2} + 10^{-2}$

$x = 10^{-12}$ $\qquad$ **pH = 1**

$\alpha_{H_2O} = \frac{10^{-12}}{\frac{1000}{18}} = 1.8 \times 10^{-14}$

$HCl \longrightarrow H^+ + Cl^-$

$10^{-8}M \qquad 10^{-8}M$

$H_2O \rightleftharpoons H^+ + OH^-$

$10^{-8}+x \qquad x$

$x(10^{-8}+x) = 10^{-14}$

$x^2 + 10^{-8}x - 10^{-14} = 0$

$$x = \frac{-10^{-8} + \sqrt{10^{-16} + 4 \times 10^{-14}}}{2}$$

$$x = \frac{-10^{-8} + 10^{-8}\sqrt{401}}{2}$$

$$x = \frac{19}{2} \times 10^{-8}$$

$$[H^+] = 10^{-8} + \frac{19}{2} \times 10^{-8}$$

$$= \frac{21}{2} \times 10^{-8}$$

$$pH = -\log\left(\frac{21}{2} \times 10^{-8}\right)$$

$$pH = -[\log 21 - 0.3 - 8]$$

$$pH = -[0.7 + 0.6 - 8.3] \quad (1.3)$$

$$pH = 7$$

$$\boxed{pH = 6.98}$$



pH of $10^{-8}M$ NaOH

$NaOH \longrightarrow Na^+ + OH^-$

$10^{-8} \qquad 10^{-8}$

$H_2O \rightleftharpoons H^+ + OH^-$

$x \qquad 10^{-8}+x$

$(10^{-8}+x)\,x = 10^{-14}$

$x = 19/2 \times 10^{-8}$

$$[OH^-] = 10^{-8} + \frac{19}{2} \times 10^{-8}$$

$pOH = 6.98 \qquad pH = 14 - 6.98 = 7.02 \qquad \boxed{pH = 7.02}$

$$-\log[H^+] = -\log\left[\frac{19}{2}\times10^{-8}\right]$$

$$= -[\log 19 \div 0.3 - 8]$$

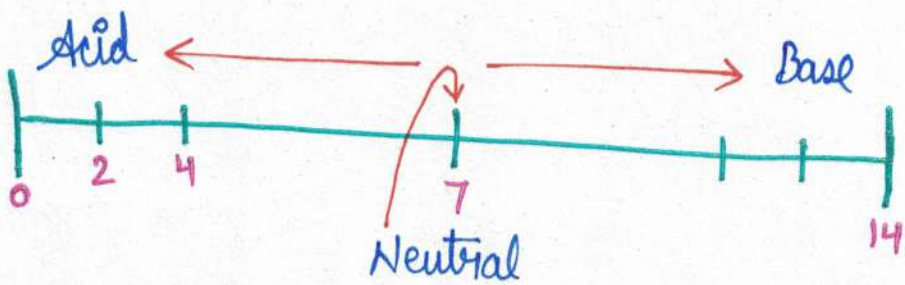
$$= \frac{0.48\times2 + 0.3 + 1.3}{2}$$

$$= \frac{0.96 + 1.6}{2}$$

$$= \frac{2.56}{2} \Rightarrow 1.28$$

$$pH = 8.3 - 1.28$$

$$\boxed{pH = 7.02}$$



If $[H^+] < 10^{-6}M$ Then consider water equilibrium catalyst are substances which can affect the rate of reaction. It cannot disturb the equilibrium.

The amount of catalyst before and after the reaction will be constant.

SALT HYDROLYSIS

Salts are of 4 kinds :-

1. Salt of strong acid, strong base

After complete neutralisation of strong acid with strong base, we'll get these type of salt.

e.g :- NaCl, K_2SO_4 etc.

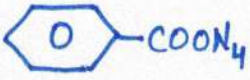
2. Salt of strong acid and weak base

After complete neutralisation of strong acid and weak base, we'll get these type of salts.

e.g:- NH_4Cl, CH_3NH_3Cl etc.

3. Salt of strong base and weak acid

After complete neutralisation of weak acid and strong base, we'll get these type of salts.

e.g:- CH_3COONa,  etc.

4. Salt of weak acid and weak base

After neutralisation of weak acid and weak base we'll get these type of salts.

e.g:- CH_3COONH_4, $HCOONH_4$ etc.

When salt is dissolved in water, the anion or cation which comes from weak acid or base will react with H_2O, the process is called spectator ion.

1. Strong Acid + Strong Base

$$NaCl \longrightarrow Na^+ + Cl^-$$

$$[H^+] = [OH^-] = 10^{-7}$$

$$pH = 7$$

2. Strong Acid + Weak Base

$$NH_4Cl \longrightarrow NH_4 + Cl^- \rightarrow \text{spectator}$$

$$NH_4^+ + H_2O \rightleftharpoons NH_3 + H_3O^+$$

t=0	c			
t=t	c(1-h)		ch	ch

$H_3O^+ \equiv H^+$

$$K_h = \frac{[NH_3][H^+]}{[NH_4^+]} = \frac{ch^2}{1-h} \qquad 1-h \simeq 1$$

$h \rightarrow$ degree of hydrolysis

$K_h \rightarrow$ hydrolysis constant

$$ch^2 = K_h$$

$$\boxed{h = \sqrt{\frac{K_h}{c}}}$$

$$NH_3 + H_2O \rightleftharpoons NH_4^+ + OH^-$$

$$K_b = \frac{[NH_4^+][OH^-]}{[NH_3]} \frac{[H^+]}{[H^+]}$$

$$K_b = \frac{K_w}{K_h} \Rightarrow \boxed{K_h = \frac{K_w}{K_b}}$$

$$K_{b_{NH_3}} \; K_{b_{NH_4^+}} = 10^{-14} = K_w$$

$$[H^+] = ch = c\sqrt{\frac{K_h}{c}} = \sqrt{cK_h}$$

$$[H^+] = \sqrt{\frac{K_w}{K_b} c}$$

$$-\log H^+ = pH = -\frac{1}{2}(\log K_w + \log c - \log K_b)$$

$$= -\frac{1}{2}(\log 10^{-14} + \log c - \log K_b)$$

$$\boxed{pH = 7 - \frac{1}{2}(\log c + pK_b)}$$

$pH < 7$

QUE. What is the pH of 0.1M NH_4Cl solution given K_b of $NH_3 = 1.8 \times 10^{-5}$
also K_b of $NH_3 = 1.8 \times 10^{-5}$
also calculate % hydrolysis of NH_4Cl

Soln:- $NH_4Cl \longrightarrow NH_4^+ + Cl^-$
0.1

$NH_4^+ \longrightarrow NH_3 + H_3O^+$

$0.1(1-h)$ $\quad 0.1h \quad 0.1h$

$$h = \sqrt{\frac{K_h}{0.1}}$$

$$K_h = \frac{K_w}{K_b} = \frac{10^{-14}}{1.8 \times 10^{-5}} = \frac{10^{-7}}{1.8}$$

$$h = \sqrt{\frac{10^{-9}}{1.8 \times 0.1}}$$

$$h = \sqrt{\frac{10^{-9}}{1.8}} = \frac{\sqrt{1.8}}{1.8} = 10^{-5}$$

$$\%h = \frac{\sqrt{18}}{18} \times 10^{-3}$$

$$-\log x = \log \sqrt{\frac{10^{-15}}{1.8 \times 10^{-5}}} = \log x$$

$$= 5.13$$

$$pH = \frac{7-1}{2}\left(pK_b - 1\right)$$

$$pH = 7 - \frac{1}{2}\left[5 - 0.26 - 1\right]$$

$$pH = 7 - \frac{1}{2}(3.74)$$

$$pH = 7 - 1.87$$

$$\boxed{pH = 5.13}$$

3. Weak Acid + Strong Base

$CH_3COONa \longrightarrow CH_3COO^- + Na^+$

$CH_3COO^- + H_2O \longrightarrow CH_3COOH + OH^-$

$c(1-h)$ $\quad ch \quad ch$

$$K_h = \frac{ch^2}{1-h} \qquad \boxed{h = \sqrt{\frac{K_h}{c}}} \qquad 1-h \simeq 1$$

$CH_3COOH \rightleftharpoons CH_3COO^- + H^+$

$$K_a = \frac{[CH_3COO^-][H^+]}{[CH_3COOH]} \quad \frac{[OH^-]}{[OH^-]}$$

$$K_a = \frac{K_w}{K_h}$$

$$\boxed{K_h = \frac{K_w}{K_a}}$$

$$[OH^-] = ch = c\sqrt{\frac{K_h}{c}} = \sqrt{cK_h}$$

$$pOH = -\log[OH^-] = \frac{-1}{2}[\log c + \log K_w - \log K_a]$$

$$pH = 14 + \frac{1}{2}(\log c - 14 + pK_a)$$

$$\boxed{pH = 7 + \frac{1}{2}(\log c + pK_a)}$$

$pH > 7$

basic solution

QUE. pH of 0.1M CH_3COONa = ?
If K_a of $CH_3COOH = 1.8 \times 10^{-5}$
Also calculate % hydrolysis

Soln:- $CH_3COONa \longrightarrow CH_3COO^- + Na^+$

0.1 0.1

CH_3COO^-	CH_3COOH	$+ OH^-$
$0.1(1-h)$	$0.1h$	$0.1h$

$$K_h = \frac{0.1(h^2)}{1-h}$$

$$h = \sqrt{\frac{K_h}{0.1}}$$

$$[OH^-] = 0.1h = 0.1\sqrt{\frac{K_h}{0.1}}$$

$$pOH = -\frac{1}{2}(\log(0.1) + \log(10^{-14}) - \log K_a)$$

$$pH = 7 + \frac{1}{2}\left(1 + 14 - 0.26\right)$$

$$= 7 + \frac{15.26}{2}$$

$$= 7 + 1.87$$

$$\boxed{pH = 8.87}$$

4. Salt of Weak Acid + Weak Base

$$CH_3COONH_4 \longrightarrow CH_3COO^- + NH_4^+$$

$$\underset{c}{NH_4^+} + \underset{c}{CH_3COO^-} + H_2O \rightleftharpoons \underset{c}{CH_3COOH} + OH^- + H^+ + NH_3$$

$t=0$	c	c
$t=t$	$c(1-h)$	$c(1-h)$

$$CH_3COOH + NH_4OH$$

$$K_h = \frac{[CH_3COOH][NH_4OH]}{[CH_3COO^-][NH_4^+]} = \frac{\overset{ch}{ch}}{\underset{c(1-h)}{c(1-h)}}$$

$$\boxed{K_h = \left(\frac{h}{1-h}\right)^2}$$

$CH_3COO\,(K_a)$

$$CH_3COOH \rightleftharpoons CH_3COO^- + H^+$$

$$K_a = \frac{[CH_3COO^-][H^+]}{[CH_3COOH]}$$

$NH_4OH\,(K_b)$

$$NH_4OH \rightleftharpoons NH_3^+ + OH^-$$

$$K_b = \frac{[NH_3^+][OH^-]}{[NH_4OH]}$$

$$K_a \cdot K_b = \frac{[CH_3COO^-][H^+]}{[CH_3COOH]} \cdot \frac{[NH_3^+][OH^-]}{[NH_4OH]}$$

$$K_a \cdot K_b = \frac{K_w}{K_h}$$

$$\boxed{K_h = \frac{K_w}{K_a \cdot K_b}}$$

$$\frac{h}{1-h} = \sqrt{K_h}$$

$$h = \sqrt{K_h} - h\sqrt{K_h}$$

$$\boxed{h = \frac{\sqrt{K_h}}{1+\sqrt{K_h}}}$$

$$K_a = \frac{[CH_3COO^-][H^+]}{[CH_3COOH]}$$

$$K_a = \frac{c(1-h)[H^+]}{ch}$$

$$[H^+] = \frac{h \cdot K_a}{1-h}$$

$$[H^+] = K_a\sqrt{\frac{K_w}{K_a K_b}} = \sqrt{\frac{K_a K_w}{K_b}}$$

$$pH = -\log[H^+] = -\tfrac{1}{2}(\log k_a + \log k_w - \log k_b)$$

$$\boxed{pH = 7 + \tfrac{1}{2}(pK_a - pK_b)}$$

1. if $K_a = K_b$ $pH = 7$ (Neutral Salt)
2. if $K_a > K_b$ $pK_a < pK_b$ $pH < 7$ (Acidic Salt)
3. if $K_a < K_b$ $pK_a > pK_b$ $pH > 7$ (Basic Salt)

In this case, K_h & pH are independent of the concentration.

BUFFER SOLUTION

Solution having constant pH is called buffer solution.

It is of 2 kind:-

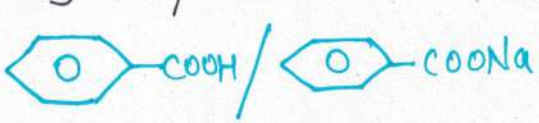
1. ACIDIC BUFFER

A Solution having weak acid with its conjugate base in significant amount.

2. BASIC BUFFER

A Solution having weak base with its conjugate acid in significant amount.

QUESTION Identify buffer:-

1. CH_3COOH / CH_3COO^- ✓ Acidic buffer
2.  ✓ Acidic buffer
3. NH_3 / NH_4Cl ✓ Basic buffer
4. CH_3NH_2 / CH_3NH_3Cl ✓ Basic buffer
5. HCl / Cl^- ✗ Not buffer
6. H_2SO_4 / HSO_4^- ✗ Not buffer
7. H_2SO_4 / SO_4^{2-} ✗ Not buffer
8. H_2S / HS^- ✓ Acidic buffer
9. HS^- / S^{2-} ✓ Acidic buffer

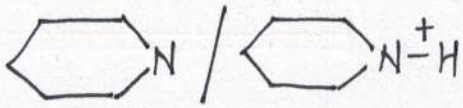
10. H_2S / S^{2-} ✗ Not Buffer

11. H_3PO_4 / HPO_4^{2-} ✗ Not Buffer

12. $H PO_4^{2-} / PO_3^{2-}$ ✓ Acidic Buffer

13. NH_2NH_2 / NH_2NH_3Cl ✓ Basic Buffer

14. $NH_2NH_2 / (NH_3NH_3)Cl_2$ ✗ Not Buffer

15.  ✓ Basic buffer

PH CALCULATION FOR ACIDIC BUFFER

CH_3COO^- / CH_3COOH

Concentration :- c_1 c_2 $c_1 = \frac{a}{V}$

moles :- a b $c_2 = \frac{b}{V}$

$$CH_3COOH \rightleftharpoons CH_3COO^- + H^+$$

$t=0$	c_2	c_1	
$t=t$	$c_2 - x$	$c_1 + x$	x

$$K_a = \frac{(c_1+x)(x)}{c_2 - x} \quad \text{(ignore } x \text{ w.r.t. } c_1 \& c_2\text{)}$$

$$K_a = \left(\frac{c}{c_2}\right)x$$

$$\{x = [H^+] \Rightarrow -\log x = -\log[H^+]\}$$

$$= pH$$

$$-\log K_a = -\log\left(\frac{c_1}{c_2}\right) - \log x$$

$$pK_a = -\log\left(\frac{c_1}{c_2}\right) + pH$$

$$pH = pK_a + \log\left(\frac{c_1}{c_2}\right)$$

$$pH = pK_a + \log\frac{[\text{salt}]}{[\text{Acid}]}$$ → Hinduson's equation

$$pH = -pK_a + \log\left[\frac{a}{b}\right]$$ → Constant

I. If H_2O is added

$$pH = pK_a + \log\left(\frac{a}{b}\right)$$

II. x mole HCl, $x \leq a, b$

$$CH_3COO^- + HCl \rightarrow CH_3COOH$$

a	x	b
a-x	[LR]	b+x

$$pH = pK_a + \log\left(\frac{b-x}{a+x}\right)$$ x is small

$$pH = pK_a + \log\left[\frac{b}{a}\right]$$

III. Y mole NaOH, $Y \ll a, b$

$$CH_3COOH + NaOH \rightarrow CH_3COO^-$$

b	Y	H
b-Y	[LR]	a+Y

$$pH = pK_a + \log\left[\frac{a+Y}{b-Y}\right]$$

Y is small

$$pH = pK_a + \log\left(\frac{a}{b}\right)$$ → Constant

QUESTION 10 mole CH_3COOH is present with 10 mole CH_3COONa in 1 lt solution. what will be

(i) the pH of this solution

(ii) if 2 lt water is added

(iii) 2 mole NaOH added

(iv) 2 mole KCL added

(i)

$$CH_3COOH \rightleftharpoons CH_3COO^- + H^+$$
$$10 \qquad\qquad 10$$

$$pH = pK_a + \log\left(\frac{a}{b}\right)$$

$$pH = -\log(1.8\times10^{-5}) + \log\left(\frac{10}{10}\right)$$

$$pH = -(0.26-5) + \log 1$$

$$pH = 5 - 0.26 = 4.74$$

(ii) If 2 lt water is added

$$CH_3COOH \rightleftharpoons CH_3COO^- + H^+$$

[acid] = $10/2 = 5$

[Salt] = 5

$$pH = pK_a + \log\left[\frac{5}{5}\right]$$

$$pH = 4.74$$

(iii) If 2 Mole HCL is added

$$CH_3COOH + HCL \rightarrow CH_3COOH$$
$$10 \qquad 2$$
$$10-2=8 \qquad 0 \qquad 10+2=12$$

$$pH = 4.74 + \log\left[\frac{12^3}{8_2}\right]$$

$$pH = 4.74 + 0.48 - 0.3$$

$$pH = 4.92$$

iv

$CH_3COO^- + HCl \longrightarrow CH_3COOH$

10	2	10
8	LR	12

$pH = 4.74 + \log\left(\frac{8}{12}\right)$

$pH = 4.74 + 0.3 - 0.48$

$pH = 4.56$

PH CALCULATION FOR BASIC BUFFER

$NH_4^+ / NH_3 \quad (K_b)$

a	b	$c_1 = \frac{a}{V}$
c_1	c_2	$c_2 = \frac{b}{V}$

$NH_3 + H_2O \rightleftharpoons NH_4^+ + OH^-$

c_2	c_1	
$c_2 - x$	$c_1 + x$	x

$[-\log x = pOH]$

$K_b = \frac{[NH_4^+][OH^-]}{[NH_3]} = \left(\frac{c_1 + x}{c_2 - x}\right) x$

$-\log K_b = -\log \frac{c_1}{c_2} - \log x$

$pK_b = -\log \frac{c_1}{c_2} + pOH$

$pOH = pK_b + \log\left[\frac{\text{salt}}{\text{Base}}\right]$

$pOH = pK_b + \log\left[\frac{a}{b}\right]$

v. Dilution $pH = 14 - pOH$

2. x mole HCL is added.

$NH_3 + HCL \longrightarrow NH_4^+$

	b	x	a
	b-x	[LR]	a+x

$$pOH = pK_b + \log\left(\frac{a+x}{b-x}\right)$$

3. Y mole NaOH is added

$NH_4 + NaOH \longrightarrow NH_3$

	a	Y	b
	a-Y	[LR]	b+Y

$$pOH = pK_b + \log\left(\frac{a-Y}{b+Y}\right)$$

QUE. 10 mole NH_3 is taken with 10 mole NH_4^+ in 1 lt soln. what will be the pH of soln

i. $K_b = 1.8 \times 10^{-5}$ → of NH_3

ii. if 2 lt H_2O is added

iii. if 2 mole HCL added

iv. 2 mole NaOH added

Soln:- i.

$$NH_3 + H_2O \rightleftharpoons NH_4 + OH^-$$

10 10

$$pOH = pK_b + \log\left(\frac{10}{10}\right)$$

$$pOH = -\log(1.8 \times 10^{-5})$$

$$pOH = 4.74$$

$$pOH = 14 - 4.74 = 9.26$$

ii. No effect on dilution

$$pH = 9.26$$

iii $NH_3 + HCL \longrightarrow NH_4^+$

$t \neq 0$	10	2	10
$t = t$	10-2 = 8		10+2 = 12

$$pOH = 4.74 + \log \frac{12/8}{3/2}$$

$$pOH = 4.74 + 0.48 - 0.3$$

$$pOH = 4.92$$

$$pOH = 14 - 4.92 = 9.08$$

iv $pOH = 4.74 + 0.3 - 0.48$

$= 5.04 - 0.48$

$= 4.56$

$pOH = 14.00 - 4.56$

$= 9.44$

QUE What is PH?

i 500 ml 0.1 M CH_3COOH $(K_a = 10^{-5})$

$CH_3COOH \longrightarrow CH_3COO^- + [H^+]$

$[H^+] = \sqrt{cK_a}$

$[H^+] = \sqrt{0.1 \times 10^{-5}}$

$[H^+] = \sqrt{10^{-6}}$

$[H^+] = 10^{-3}$

$pH = -\log(10^{-3}) = 3$

ii 500 ml 0.1 M COOH + 100 ml 0.1 M NaOH

$CH_3COOH + NaOH \longrightarrow CH_3COO^-$

millimole	50	10	
	40	[LR]	10

This is a buffer solution

$$pH = pKa + \log\left(\frac{10}{40}\right)$$

$$pH = -\log(10^{-5}) + \log(1/4)$$

$$pH = 5 + 0 - 2(0.3)$$

$$pH = 5 - 0.6$$

$$pH = 4.4$$

(iii) 500 ml 0.1M CH_3COOH + 300 ml 0.1M NaOH

$$CH_3COOH + NaOH \rightarrow CH_3COO^-$$

50 30

50-30 [LR] 30

This is a buffer solution

$$pH = 5 + \log\left(\frac{3\phi}{2\phi}\right)$$

$$pH = 5 + \log 3 - \log 2$$

$$pH = 5 + 0.48 - 0.3$$

$$pH = 5.18$$

(IV) 500 ml 0.1M CH_3COOH + 500 ml 0.1M NaOH

$$CH_3COOH + NaOH \rightarrow CH_3COONa$$

50 50 50

$$[CH_3COONa] = \frac{50}{1000} = \frac{5}{100}$$

Salt hydrolysis

$$pH = 7 + \frac{1}{2}[\log C + pKa]$$

$$pH = 7 + \frac{1}{2}\left[\log\left(\frac{5}{100}\right) + 5\right]$$

$$pH = 7 + \frac{1}{2}[0.7 - 2 + 5]$$

$$pH = 7 + \frac{1}{2}(3.7)$$

$$pH = 7 + 1.85$$

$$pH = 8.85$$

Y. 500 ml 0·1M CH_3COOH + 700 ml 0·1M NaOH

$CH_3COOH + NaOH \longrightarrow CH_3COO^-$

50	70	
[LR]	70 − 50 = 20	50

Weak base + strong base

$[OH^-]$ governed from strongbase

$$[OH^-] = \frac{20}{120} = \frac{1}{60}$$

$$pOH = -\log\left(\frac{1}{60}\right) = \log 60 = \log 2 + \log 3 + 1$$
$$= 0.3 + 0.48 + 1$$
$$= 1.78$$

$$pH = 14 - 1.78$$

$$pH = 12.26$$

WEAK POLYPROTIC ACID

H_2S (Ka_1, Ka_2) $\quad Ka_1 >> Ka_2$

$Ka_1 \rightarrow$ $H_2S \rightleftharpoons H^+ + HS^-$

$c-x \quad x+y \quad x$

$HS^- \rightleftharpoons H^+ + S^{2-}$

$Ka_2 \rightarrow$ $x-y \quad x+y \quad y$

We can ignore
x wrt c
y wrt x

$$Ka_1 = \frac{[H^+][HS^-]}{[H_2S]} = \frac{(x+y)(x)}{c-x}$$

$Ka_1 >> Ka_2$

$$Ka_1 = \frac{x^2}{c} \qquad x = \frac{c}{\sqrt{cKa_1}}$$

$[H^+]$ will be governed from Ist step

$$Ka_2 = \frac{[H^+][S^{2-}]}{[HS^-]} = \frac{(x+y)(y)}{(x-y)^{\circ}} = Y$$

$$Y = Ka_2 = [S^{2-}]$$

$$\alpha_{H_2S} = \frac{x}{C} \quad \% \alpha_{H_2S} = 0.1\%$$

$$\underset{C-x}{H_3PO_4} \rightleftharpoons \underset{x+y+z}{H^+} + \underset{x}{H_2PO_4^-}$$

$$Ka_1 = \frac{(x+y+z)x}{C-x} = \frac{x^2}{C}$$

$$\underset{x-y}{H_2PO_4^-} \rightleftharpoons \underset{x+y+z}{H^+} + \underset{y}{HPO_4^{2-}}$$

$$Ka_2 = \frac{(x+y+z)y}{x-y} \simeq Y = Ka_2$$

$$\underset{x-y}{HPO_4^{2-}} \rightleftharpoons \underset{x+y+z}{H^+} + \underset{y}{PO_4^{3-}}$$

$$Ka_3 = \frac{(x+y+z)z}{y-z} \simeq \frac{xz}{y} = Ka_3$$

where H^+ is governed from strong acid (outside).

$$\underset{c_1}{HCL} + \underset{c_2}{H_2S}$$

$$\underset{c_2-x}{H_2S} \rightleftharpoons \underset{c_1+2x}{2H^+} + \underset{x}{S^{2-}}$$

$$Ka = Ka_1 \cdot Ka_2 = \frac{[H^+]^2[S^{2-}]}{[H_2S]}$$

$Ka_1, Ka_2, [H^+]$ $[H_2S]$ will be given $[S^{-2}]$ can be calculated.

$\downarrow c_1$ $\downarrow c_2$

SOLUBILITY AND SOLUBILITY PRODUCT

In this section, we'll study about sparingly soluble salts.

$AgCl_{(s)} \rightleftharpoons Ag^+_{(aq)} + Cl^-_{(aq)}$

S $\frac{mole}{lt}$ S S

$K_{sp} = [Ag^+][Cl^-] = S.S = S^2$

$S = \sqrt{K_{sp}}$

K_{sp} = solubility product

K_{sp} only depend on temperature.

on addition of solid at a particular temperature will have no effect on solubility.

solvent $\longrightarrow H_2O$

e.g:- $PbCl_{2(s)} \rightleftharpoons Pb^{+2}_{(aq)} + 2Cl^-_{(aq)}$

$K_{sp} = [Pb^{+2}][Cl^-]^2 = S.(2S)^2 = 4S^3$

$S^3 = \frac{K_{sp}}{4}$

$S = \left(\frac{K_{sp}}{4}\right)^{1/3}$

S is solubility of $PbCl_2$

Generalise $A_xB_y \rightleftharpoons xA^{y+} + yB^{x-}$

x^3 y^s

$K_{sp} = (xs)^x(ys)^y = x^x y^y s^{x+y}$

e.g:- $Na_3AlF_6 \rightleftharpoons 3Na^+ + AlF_6^{3-}$

$K_{sp} = (3s)^3(s)$

$K_{sp} = 27s^4$

$AgCl_{(s)} \rightleftharpoons Ag^+_{(aq)} + Cl^-_{(aq)}$

Ionic product $I.P = [Ag^+][Cl^-]$

if $I.P = K_{sP}$ saturated solution

$I.P < K_{sP}$ unsaturated solution

$I.P > K_{sP}$ supersaturated solution (ppt occur)

{ Remember Q_c, k_c !! }



QUE. AgCl ($K_{sp} = 10^{-10}$)

(i) What will be solubility in water

(ii) what will be solubility in 0.1M KCl soln

(iii) What will be solubility in 0.1M $AgNO_3$ soln

Soln:- (i) $AgCl_{(s)} \rightleftharpoons Ag^+_{(aq)} + Cl^-_{(aq)}$

s s

$K_{sp} = s^2$

$s = \sqrt{K_{sP}}$

$= \sqrt{10^{-10}}$

$= 10^{-5}$

(ii) $KCl \longrightarrow K^+ + Cl^-$

0.1 0.1 0.1

$0.1 + s'$

$AgCl \rightleftharpoons Ag^+ + Cl^-$

s' $0.1 + s'$

$K_{sp} = s'(0.1 + s)$

$10^{-10} = 10^{-1} s'$

$s' = 10^{-9}$ mol/lt

(iii) $AgCl \rightleftharpoons Ag^+ + Cl^-$

s'' $0.01 + s''$

$10^{-10} = 10^{-2} s''$

$s'' = 10^{-8}$ mol/lt

$PbCl_2$ ($K_{sp} = 4 \times 10^{-12}$)

What will be solubility

(i) in H_2O

(ii) in 0.01 M HCl

(iii) in 0.01 M $Pb(NO_3)_2$

(i) $PbCl_2 \rightleftharpoons Pb^+ + 2Cl^-$

s s $2s$

$K_{sp} = s.(2s)^2$

$\frac{4 \times 10^{-12}}{4} = s^3$ $\quad s = 10^{-4}$ mol/lt

(ii) $PbCl_2 \rightleftharpoons Pb^{+2} + 2Cl^-$

s' $10^{-2} + 2s'$

$K_{sp} = (10^{-2} + 2s)^2 s'$

$\frac{4 \times 10^{-12}}{10^{-2}} = s'$ $\quad s' = 4 \times 10^{-8}$

iii

$PbCl_2 \rightleftharpoons Pb^{+2} + 2Cl^-$

s'' $\quad 10^{-2} + 2s''$

$4 \times 10^{-12} = s''(10^{-2} + 2s'')^2$

$PbCl_2 \rightleftharpoons Pb^{+2} + 2Cl^-$

$S + 10^{-2} \quad 2S$

$Pb(NO_3)_2 \longrightarrow Pb^{+2} + 2NO_3^{2-}$

$4 \times 10^{-12} = (S + 10^{-2})(2S)^2$

$4 \times 10^{-12} = 10^{-2} \times 4S^2$

$10^{-10} = S^2$

$S = 10^{-5}$

Here, solubility is decreasing due to common ion.

SIMULTANEOUS SOLUBILITY

$AgCl_{(s)} \rightleftharpoons Ag^+_{(aq)} + Cl^-_{(aq)}$

$s_1 + s_2 \quad s_1$

$Ksp_1 = (s_1 + s_2)s_1$

$AgCN_{(s)} \rightleftharpoons Ag^+_{(aq)} + CN^-_{(aq)}$

$s_1 + s_2 \quad s_2$

$Ksp_2 = (s_1 + s_2)s_2$

$Ksp_1 + Ksp_2 = (s_1 + s_2)(s_1 + s_2)$

$= (s_1 + s_2)^2$

$s_1 + s_2 = \sqrt{Ksp_1 + Ksp_2}$

SELECTIVE – PRECIPITATION

QUE In a solution $[I^-] = [Br^-] = [Cl^-] = 0.01M$
$AgNO_3$ is added slowly.

(i) Which ion will precipitate first?

(ii) What will be % precipation of first ion, when 2nd will start precipitating?

(iii) What will be % precipitation of other ions when last ion will start precipitating.

(Given :- Ksp of AgCl $= 10^{-10}$, Ksp of AgBr $= 10^{-13}$)
Ksp of AgI $\longrightarrow 10^{-16}$

Soln:- for AgCl $[Ag^+][Cl^-] = K_{sp}$

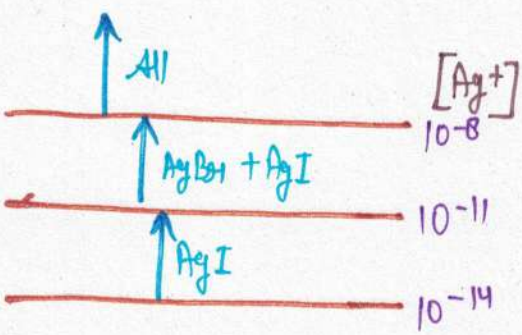
$$[Ag^+] = \frac{10^{-10}}{10^{-2}} = 10^{-8}$$

for AgBr $[Ag^+][Br^-] = 10^{-13}$

$$[Ag^+] = \frac{10^{-13}}{10^{-2}} = 10^{-11}$$

for AgI $[Ag^+][I^-] = 10^{-16}$

$$[Ag^+] = \frac{10^{-16}}{10^{-2}} = 10^{-14}$$



(i) AgI will precipitate first.

(iii) When AgBr will start precipitating.
$[Ag^+] = 10^{-11}$

for AgI = $[Ag^+][I^-] = 10^{-16}$

$$[I^-] = \frac{10^{-16}}{10^{-11}} = 10^{-5}$$

$$\% \text{ remain} = \frac{10^{-5}}{10^{-2}} \times 100 = 0.1\%$$

$$\% \text{ ppt} = 100 - 0.1$$
$$= 99.9\%$$



$[Ag^+] = 10^{-8}$

for AgI $[Ag^+][I^-] = 10^{-16}$

$$[I^-] = \frac{10^{-16}}{10^{-8}} = 10^{-8}$$

$$\% \text{ remain} = \frac{10^{-8}}{10^{-2}} \times 100 = 10^{-4}\%$$

$$\% \text{ ppt} = 100 - 0.0001$$
$$= 99.9999\%$$

for AgBr $[Ag^+][Br^-] = 10^{-13}$

$$[Br^-] = \frac{10^{-13}}{10^{-8}} = 10^{-5}$$

$$\% \text{ remain} = \frac{10^{-5}}{10^{-2}} \times 100 = 10^{-1}$$

$$\% \text{ ppt} = 100 - 0.1 = 99.9\%$$

THEORY OF INDICATORS

Indicators are organic substances which have different colours in ionised and unionised form.

phenolphthalein :- slightly basic medium

$$HPh \rightleftharpoons H^+ + Ph^-$$

colourless pink

$$K_a = \frac{[H^+][Ph^-]}{[HPh]} \Rightarrow -\log K_a = -\log [H^+] - \log \frac{[Ph^-]}{[HPh]}$$

$$CH_3COOH + NaOH \longrightarrow PH = 7 + \frac{1}{2}(\log c + pK_a)$$

range = 8.3 to 10

$$\frac{[Ph^-]}{[HPh]} \geq 10 \quad \text{pink}$$

$$\frac{[Ph^-]}{[HPh]} \leq 10 \quad \text{colourless}$$

Methyle orange :- Acidic medium

$$MeOH \rightleftharpoons Me^+ + OH^-$$

Yellow red

eg:- NH_4Cl

$$pH = 7 - \frac{1}{2}(\log c + pK_b)$$

$$K_b = \frac{[Me^+][OH^-]}{[MeOH]}$$

$$pOH = pK_b + \log \frac{[Me^+]}{[MeOH]}$$

$pOH = pK_b + 1$

pH range :- 3 to 5

if $\frac{[Me^+]}{[MeOH]} \geq 10$ red

$\frac{[Me^+]}{[MeOH]} \leq 10$ Yellow

BEFFER ACTION

$CH_3COOH \rightleftharpoons CH_3COO^- + H^+$ (weakly ionise)

$CH_3COONa \rightleftharpoons CH_3COO^- + Na^+$ (highly ionise)

- If small amount of acid is added in above bffer the following reasn will occur :-

$CH_3COO^- + H^+ \rightleftharpoons CH_3COOH$ $\quad K_a' = \frac{1}{K_a} = 10^5$

Hence the added HCl will be almost completely furnished and pH change would not occur.

as $(K_a \gtrsim 10^3)$

- If small amount of base is added in above buffer then

$CH_3COOH + OH^- \longrightarrow CH_3COO^- + H_2O$ $\quad \frac{1}{K_H} = \frac{K_a}{K_w} = \frac{10^{-5}}{10^{-14}} = 10^{+9}$

Hence the added base will be alive finished & pH change would not occure.

PH of ACIDIC BUFFER

(as $K_a > 10^3$)

$$pH = pK_a + \log\left[\frac{salt}{acid}\right]$$

$$pH = pK_a + \log\left[\frac{\text{Conjugate base}}{acid}\right]$$

Que. $pH = ?$

a. 0.1M CH_3COOH + 0.1M CH_3COONa $(pK_a = 4.74)$

$$pH = pK_a + \log\left(\frac{0.1}{0.1}\right)$$

b. equal volume of 0.1M CH_3COOH + 0.05 M NaOH

$$CH_3COOH + NaOH \longrightarrow CH_3COONa + H_2O$$

t=t	0.1V	.005V		
t=0	0.10V - 0.05x	0		0.05 x
	0.05V			0.05 Y

$$pH = pK_a + \log\left(\frac{0.05V}{0.05V}\right) = 4.74$$

Range of Buffer:- $\frac{1}{10} \leq \left[\frac{CH_3COONa}{CH_3COOH}\right] \leq 10$

i.e:- $pK_a - 1$ to $pK_a + 1$

By using an acidic Buffer, soln of pH from $pK_a - 1$ to $pK_a + 1$ can be formed.

★ In any type of 2 solutions mixtures.

1. ★ pH difference b/w then is 1, then pH of resulting mixture will be = lower solutions pH + 0.26
2. ★ pH difference is 2, then pH of resulting soln = Lower pH + 0.29
3. ★ pH difference > 2, then pH of resulting soln = Lower pH + 0.3

BUFFER CAPACITY (ϕ)

Power of buffer to resist change in its pH on addition of a small amount of an acid or a base.

$$\phi = \frac{\text{no. of moles of acid/base added in 1lt soln}}{\text{change in pH}}$$

$$\phi = \frac{d(n)}{d(pH)}$$

for max^n buffer capacity (∞), pH change should be zero

Acidic buffer :- $CH_3COOH + CH_3COONa$

moles a s

$$\phi = \frac{2.303(as)}{a+s}$$

Basic buffer :- $\phi = \frac{2.303(bs)}{b+s}$

For max^n buffer capacity

[Acid] = [salt]

or

[Base] = [salt]

QUESTION Among the following, which soln^n has max^n buffer capacity

(i) 6 gm CH_3COOH + 82 gm CH_3COONa

(ii) 6.00 gm CH_3COOH + 82 gm CH_3COONa

(iii) 60 gm CH_3COOH + 8.2 gm CH_3COONa

(iv) 6 gm CH_3COOH + 8.2 gm CH_3COONa → Moles are equal

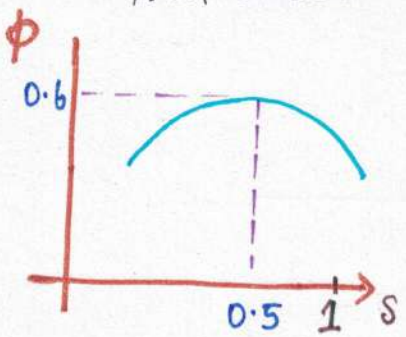
$\frac{6}{10} < \frac{82}{82}$

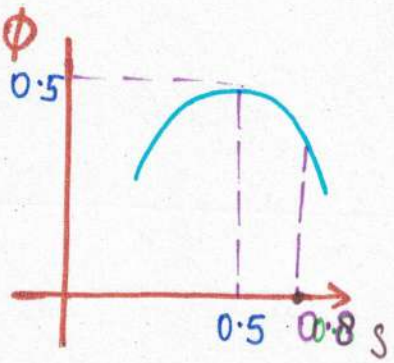
$\frac{6.00}{60} > \frac{82}{82}$

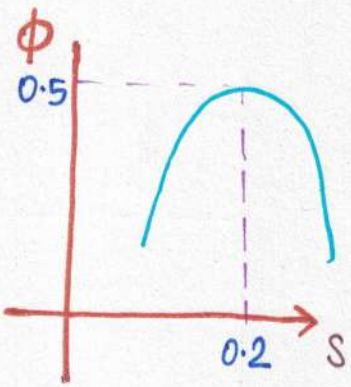
$\frac{60}{60} > \frac{82}{82}$

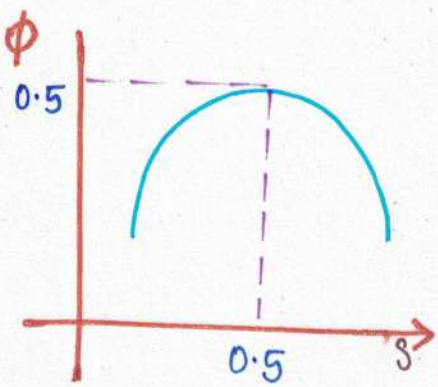
$\frac{6}{60} = \frac{8.2}{82}$

QUESTION In an acidic buffer, acid (a) and salt (s) has a+s=1. If a graph of buffer capacity will be salt amount is polluted then correct occure.









SOLUTION :- $\left[\phi = 2.303 \frac{as}{ats} = \frac{2.303 \times 0.25}{1} = 0.57575\right]$

INDICATORS AND TITRATIONS

Solutions which are used to detect the end point (equivalence pt) in acid base tibration.

Indicator		pH range		colour change
Methyle orange	→	3.2 - 4.5	→	Yellow to orange
Methyle red	→	4.4 - 6.5	→	Yellow to red
Litmus	→	5.5 - 7.5	→	red to blue

Phenol red → 6.8 - 8.4 → red to Yellow

Phenolphthalein → 8.3 - 10.5 → colourless to pink

1. OSTIVALD THEORY

Indicators are soln of weak acid or weak base which will ionise less in one medium than in other. Both ionise and unionised forms have different colours.

▷ Phenolphalein is a weak acid which ionise as

$$HPh \rightleftharpoons H^+ + Ph^-$$

unionised form → colourless

ionised form → pink

In acidic medium HPh is colourless
In Basic medium HPh is pink

▷ Methyl orange is a weak base which ionise as

$$MeOH \rightleftharpoons Me^+ + OH^-$$

unionised form → yellow

ionised form → Orange

In acidic medium Me(OH) is Orange
In Basic medium Me(OH) is yellow

▷ Usually one form which is present in Large percentage (>50%) will show its colour. But according to human eye sensitivity, one form is 10 times higher than the other, then the solution will show its colour.

$$\underset{1-\alpha}{HIn} \rightleftharpoons \underset{\alpha}{H^+} + \underset{\alpha}{In^-}$$

$\left[\frac{In^-}{HIn}\right] > 10$ (colour of In^- will be there)

$$\frac{\alpha}{1-\alpha} \geq 10$$

$$11\alpha \geq 10$$

$$\alpha \Rightarrow \frac{10}{11}$$

$$\alpha \geq 0.91$$

% ionisation $\geq$ 91 %

$\left[\frac{In^-}{HIn}\right] \leq \frac{1}{10}$ (colour of HIn will be there)

$$\frac{\alpha}{1-\alpha} \leq \frac{1}{10}$$

$$\alpha \leq \frac{1}{11}$$

$$\alpha \leq 0.091$$

% ionisation $\leq$ 9.1 %

PH OF INDICATOR SOLUTION

Solution of indicator will act as buffer.

Hence

PH of acid indicator $pH = pK_a + \log\left[\frac{IN^-}{HIN}\right]$

or. $pH = pK_a + \log\left[\frac{\alpha}{1-\alpha}\right]$

pH of basic indicator $pH = pK_a + \log\left[\frac{\text{conj. acid}}{\text{base}}\right]$

QUE. Calculate the pH of acid indicator soln which ionise to 30% in solution ($pK_{In} = 4.7$)

Soln:- $pH = pK_a + \log\left(\frac{\alpha}{1-\alpha}\right)$

$pH = 4.7 + \log\left(\frac{0.3}{0.7}\right)$

$\alpha = 0.3$

$pK_a = 4.7$

$4.7 + 0.4.8 - 1$

$-.84 + 1$

$= 5.18 - 0.84 = 4.34$

$pK_a = 4.34$

QUE. An Acid indicator show (blue) colour in acidic form and red in basic form. Blue colour will appear only when acidic form is 10 times higher than basic form and red colour will appear only when basic form is 5 times greater than acidic form what will be pH range of indicator ? ($pK_{In} = 5$)

Soln:- $HIn \rightleftharpoons In^- + H^+$

acidic — basic

Blue colour $pH = pK_a + \log\left[\frac{In^-}{HIn}\right]$

$\left[\frac{HIn}{In^-}\right] > 10$

$pH = 5 + \log(1/10)$

$pH = 5 - 1 = 4$

$\boxed{pH = 4}$

Red colour $pH = 5 + \log 5$

$\left[\frac{In^-}{HIn}\right] > 5$

$pH = 5 + 0.699$

$pH = 5.699$

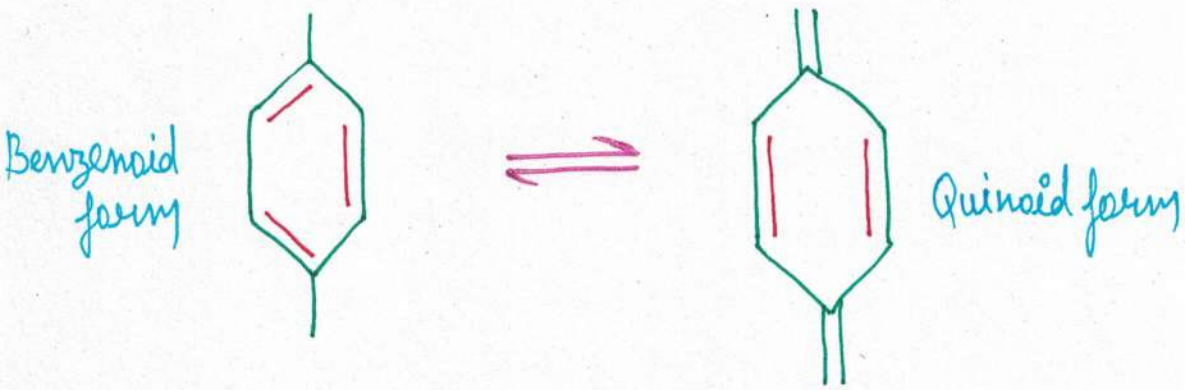
$\boxed{pH = 5.699}$

pH of indicator is of range 4 to 5.422 in the Question.

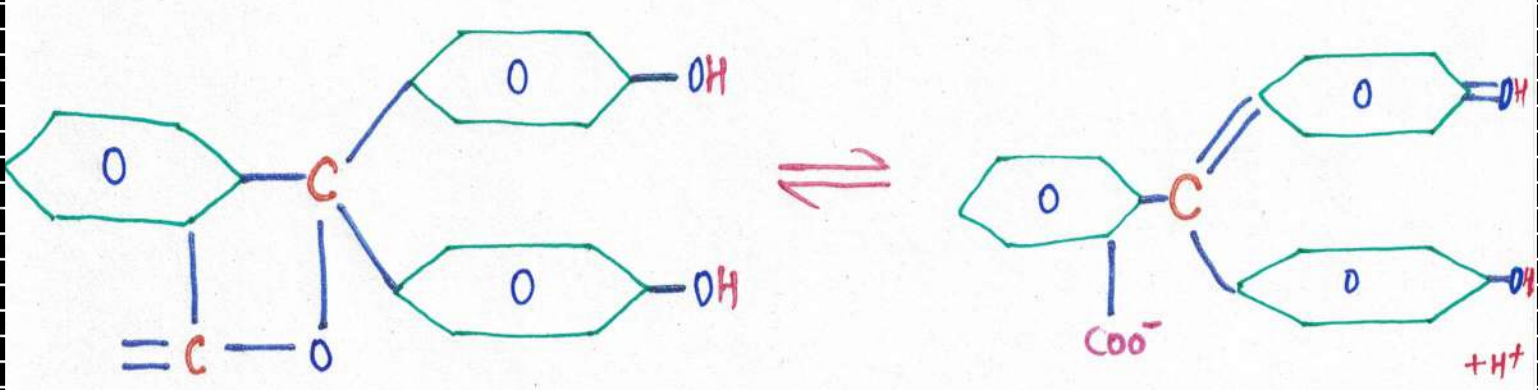
2. QUINOID THEORY

Indicators are soln of weak organic acid or weak organic base.

Indicators exist in the soln either in quinoid form or in benzenoid form in which quinoid form is dark in colour.



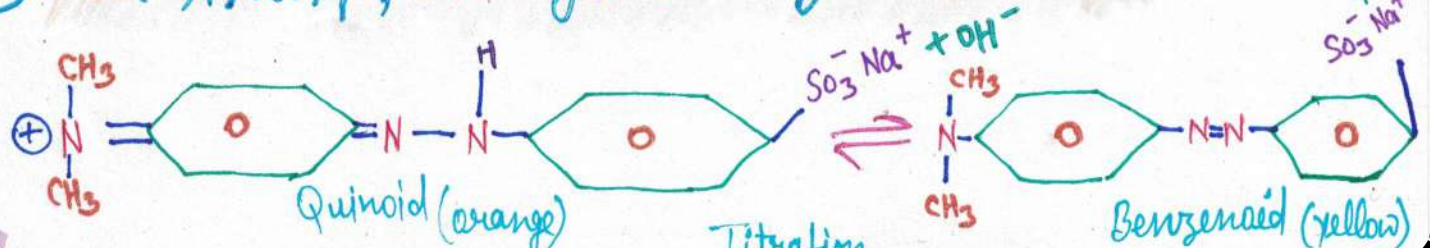
○ Phenolphthalein indicator



Benzenoid form (colourless)

Quinoid form (pink)

○ Similarly, methyl orange indicator



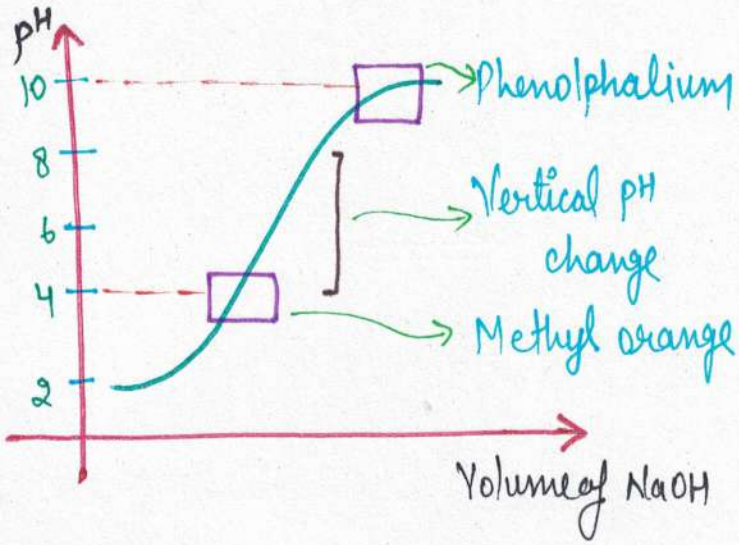
Quinoid (orange)

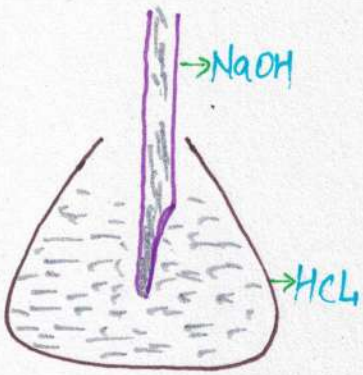
Titration

Benzenoid (yellow)

- The process in which end point is deducted by using an indicator.
- Selection of indicator is based on a vertical pH change in a titration.

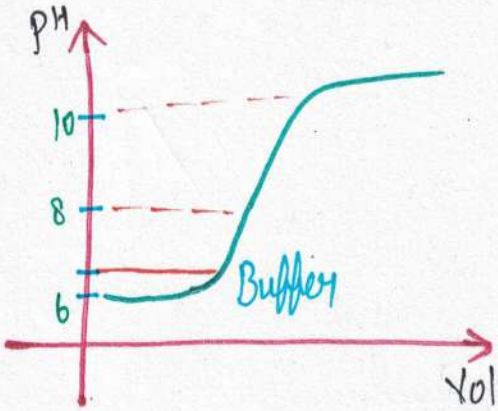
1. Titration of a Strong Acid and Strong Base

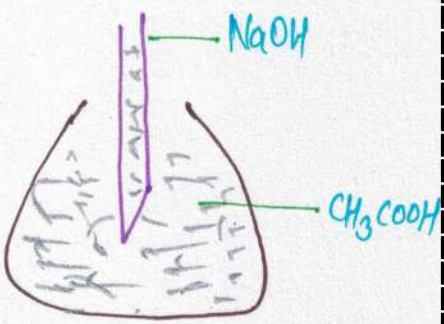




Vertical pH change is from 4 to 10

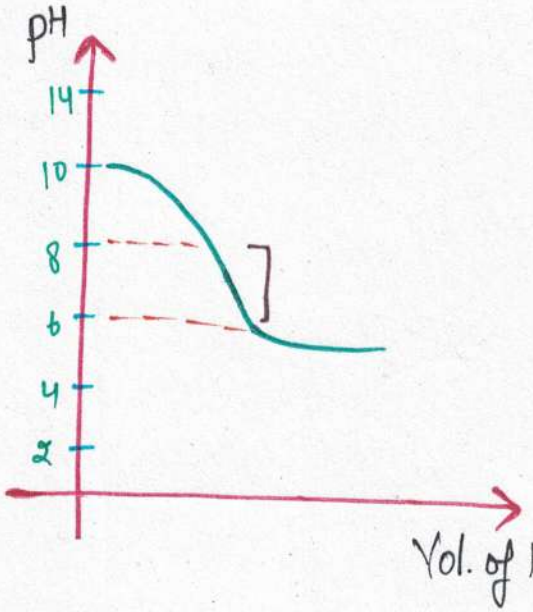
2. Titration of Weak Acid with Strong Base

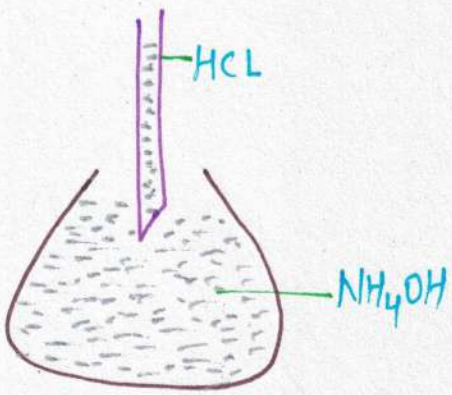




Vertical pH change is from 7 to 10

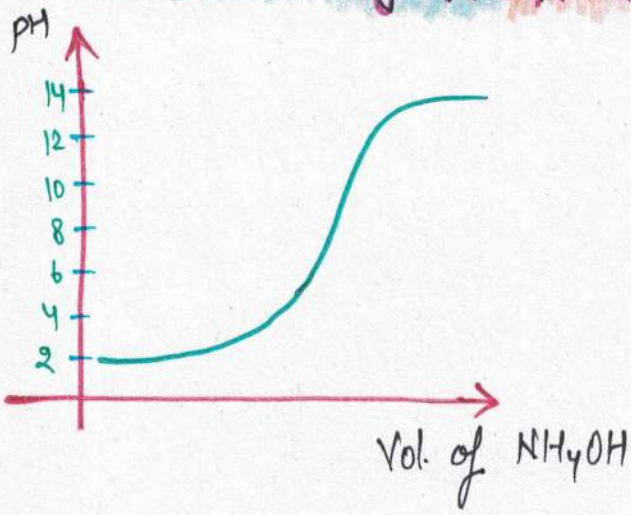
3. Titration of Weak base and strong acid

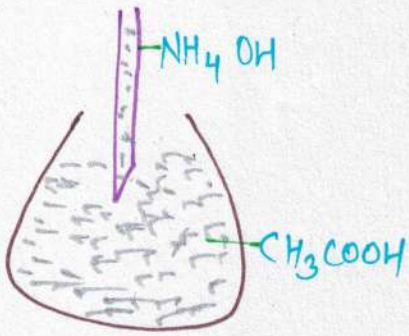




Vertical pH change is from 10 to 7

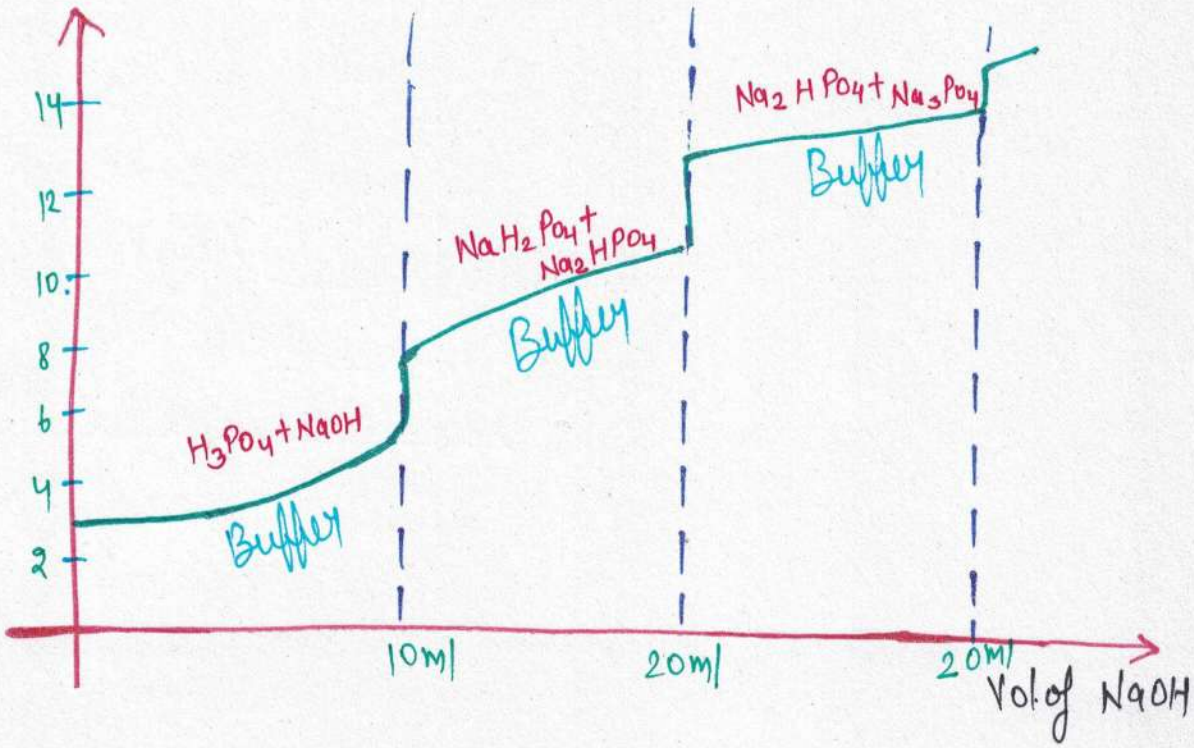
4. Titration of Weak acid and Weak base





No Vertical pH change here

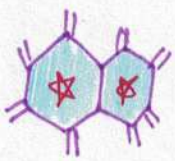
5. Titration of Weak Polyprotic acid and strong base



No. of Buffer formed will depend on polyprotic index of weak acid.

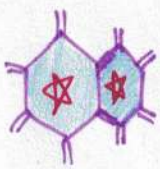
IONIC EQUILIBRIUM

SHORT NOTES



"P" operator take (-log) of the variables.

eg:- $pH = -\log(H^+)$ $\quad pKa = -\log(Ka)$



At 25°C, the $Kw = 10^{-14}$

$H_2O \rightleftharpoons H^+ + OH^-$

$Kw = [H^+][OH^-] = 10^{-14}$ ——— (1.)

Taking -log on both side of (1.)

$pH + pOH = 14$

$pH = 7 \rightarrow$ Neutral

$pH < 7 \rightarrow$ Acidic

$pH > 7 \rightarrow$ Basic

pH Calculation:- $pH = -\log[H^+] = -\log\left(\frac{\text{moles of } H^+}{\text{Volume}}\right)$

e.g:- 300ml of 0.1M HCL

millimoles of $(H^+) = 0.1 \times 300 = 30$, Volume = 300

$$pH = -\log\left(\frac{30}{300}\right) = -\log(10^{-1}) = 1$$

For strong acids:- Calculate pH of 300ml 0.3M H_2SO_4

millimoles of $H^+ = 300 \times 0.3 \times 2 = 180$

Volume = 300 ml ($2H^+$ present)

$\therefore [H^+] = \frac{180}{300}$, $pH = -\log\left(\frac{180}{300}\right)$

pH calculation for Weak Acid:-

$$CH_3COOH \rightleftharpoons CH_3COO^- + H^+$$

t=0	C			
t=t	1-α	cα	cα	

$$K_a = \frac{c\alpha^2}{1-\alpha} \Rightarrow \alpha = \sqrt{\frac{K_a}{c}} \Rightarrow [H^+] = c\alpha = \sqrt{K_a c}$$

$$pH = -\log\left(\sqrt{K_a c}\right)$$

$$pH = \frac{1}{2}(-\log c + pK_a)$$

pH Calculation for (strong + weak) Acid

- Here, pH will be governed by the strong acid only.
- Thus, only take $[H^+]$ of strong acid as the $[H^+]$ of the weak acid will be ignored.

pH Calculation for (strong + weak) Base

▷ Here, pH will be governed by the strong base only as the $[\bar{O}H]$ coming from weak base can be ignored.

Weak acid + Weak acid

(C_1, Ka_1) (C_2, Ka_2)

e.g:- CH_3COOH eg:- CCl_3COOH

$$CH_3COOH \rightleftharpoons CH_3COO^- + H^+$$

t=t $C_1 - x$ x $x+y$

$$CCl_3COOH \rightleftharpoons CCl_3COO^- + H^+$$

t=t $C_2 - y$ y $y+x$

$$x+y = [H^+] = \sqrt{C_1 Ka_1 + C_2 Ka_2}$$

$$pH = -\log(x+y)$$

Salt hydrolysis

SA = Strong acid
WA = Weak acid

a. Salt of SA + SB

NaCl
$NaOH + HCl$
(SB) (SA)

K_2SO_4 etc
$KOH + H_2SO_4$

b. Salt of SA + WB

NH_4Cl
$NH_3 + HCl$
(WB) (SA)

CH_3NH_3Cl

c. Salts of SB + WA

$CH_3COO^-\ Na^+$, 

CH_3COO^- → (SB) Na^+ → (WA)

$NaOH + CH_3COOH$

d. Salts of WA + WB

CH_3COONH_4, $HCOONH_4$

CH_3COONH_4 → CH_3COOH (WA) + NH_3 (WB)

Here, instead of K_w, K_a, K_b; We need to use K_H (hydrolysis) as to get the correct answer.

e.g:- $NH_4Cl \longrightarrow NH_4^+ + Cl^-$ (No role)

↓ Now hydrolyse etc.

$NH_4^+ + H_2O \rightleftharpoons NH_3 + H_3O^+$

$t=0$	c		
$t=t$	$c(1-h)$	ch	ch

$$K_h = \frac{(ch)^2}{c(1-h)} = (ch^2) \Rightarrow \boxed{h = \sqrt{\frac{K_h}{c}}}$$

$\boxed{K_h = \frac{K_w}{K_b}}$ → In Case of base

$\boxed{K_h = \frac{K_w}{K_a}}$ → In Case of acid

$$\boxed{pH = 7 - \tfrac{1}{2}(\log K_c + pK_b)}$$

★ In Case of Salts of WB + SA

$$pH = 7 - \frac{1}{2}(\log C + pK_b)$$

★ In Case of salts of WA + SB

$$pH = 7 + \frac{1}{2}(\log C + pK_a)$$

★ Salts of Weak acid + Weak base

$$K_h = \left(\frac{h}{1-h}\right)^2 \qquad K_h = \left(\frac{K_w}{K_a \cdot K_b}\right)$$

$$pH = 7 - \frac{1}{2}(pK_a - pK_b)$$

Buffer Solution

A Solution having constant pH is called a buffer solution.

(i) **Acidic buffer** :- A solution having weak acid along with the conjugate base in significant amount.

eg:- CH_3COOH / CH_3COO^- → (acidic buffer)

(ii) **Basic buffer** :- A solution having weak base along with the conjugate acid in significant amount

eg:- NH_3 / NH_4Cl → (Basic buffer)

pH Calculations for acidic buffer

$$pH = pK_a + \log\left(\frac{[salt]}{[Acid]}\right)$$

pH Calculations for basic buffer

$$pOH = pK_b + \log\left(\frac{[salt]}{[Base]}\right)$$

$$pH + pOH = 14$$

Weak polyprotic acid (H_3PO_4)

$$H_3PO_4 \rightleftharpoons H_2PO_4^- + H^+$$

Ka_1	t=t	c-x	x	x+y+z

$$H_2PO_4^- \rightleftharpoons HPO_4^{-2} + H^+$$

Ka_2	t=t	x-y	y	y+x+z

$$HPO_4^{-2} \rightleftharpoons PO_4^{-3} + H^+$$

Ka_3	t=t	y-z	z	z+x+y

$$Ka_1 = \frac{(x+y+z)(x)}{(c-x)} = \frac{x^2}{c}$$ — (1.)

$$Ka_2 = \frac{(x+y+z)(y)}{x-y}$$

$$Ka_2 = y \quad \text{—(2.)}$$

$$Ka_3 = \frac{(x+y+z)(z)}{(y-z)} = \frac{xz}{y} \quad \text{—(3)}$$

Calculate x, y, z from (1) (2) (3); and

$[H^+] = x+y+z$ then $pH = -\log(x+y+z)$

Solubility and Solubility prod

eg:- $AgCl_{(s)} \rightleftharpoons Ag^+ + Cl^-$ ksp

s s

$ksp = s^2$

eg:- $PbCl_2 \rightleftharpoons Pb^{+2} + 2Cl^-$

s 2s

$ksp = (s)(2s)^2 = 4s^3$